

Speaker 4 of 5 — Data Analysis 1: Demand vs Competition

Career Compass SG · Module 1 Assignment · Group 7

Your time: up to 3 minutes · You present the analysis behind the opportunity map

Your job in one sentence

Show the dimension nobody else measures — competition — and prove that "lots of jobs" and "easy to get a job" are completely different things. Your section is the analytical heart of the talk, and it sets up the Recommender demo that follows you.

Where you sit in the flow

	Speaker	Topic
	1	Dashboard overview
	2	Project technology stack
	3	Data cleaning
→ YOU	4	Data analysis 1 — Demand vs Competition
	5	Data analysis 2 — Career Recommender

Speaker 3 hands you: "...what the data tells us about demand and competition." **Your handover to Speaker 5:** "So the map shows the whole market. But how do you turn a map into advice for one person? That's the Career Recommender."

Style rule: say each finding as a plain sentence first, then point at the chart as proof. Never say "as you can see here".

Script

Part 1 — The question and the metric (0:00 – 0:40)

(Slide: the Demand vs Competition page)

My part answers one question: **where are the good odds?**

Finding where the jobs are is easy — count postings. Nobody measures the other half: **how many people are fighting for each seat?**

We measured it. Our data records how many applications each posting received and how many seats it offered. Divide one by the other and you get a simple, powerful number: **applicants per seat**. One means you're practically alone. Ten means you're in a queue.

Part 2 — The opportunity map (0:40 – 1:40)

(Slide: the four-quadrant bubble map)

We put every job category onto this one map.

- **Up** means more postings — more demand.
- **Right** means more applicants per seat — more competition.
- Bubble **size** is the number of open seats, and **colour** is pay.

Read it as four quadrants. **Top-left is the sweet spot**: plenty of jobs, hardly any queue. **Top-right is the trap**: plenty of jobs, but everyone knows it.

And the spread shocked us. **Competition differs by a factor of sixteen**. Personal Care has **0.7 applicants per seat** — fewer applicants than seats! F&B has **1.0**. But Social Services has **11.3** — more than eleven people per seat.

Here is the headline: **Information Technology is the biggest and best-paid category — 140,000 postings, \$6,500 median — but it sits mid-right on this map, with about 5 applicants per seat**. Meanwhile F&B and Customer Service sit top-left: huge volume, almost no queue. Two categories can offer similar numbers of jobs and completely different chances of getting one. **No job website will ever tell you that.**

Part 3 — The surprise finding (1:40 – 2:30)

(Slide: hard-to-fill bar chart)

One more finding from this page, and it surprised us.

Some job ads get **reposted** — the employer couldn't fill the seat the first time. You might guess those are elite, demanding roles. The data says the opposite.

Reposted jobs pay a median of \$3,700. Jobs filled first time pay \$3,850.

So hard-to-fill jobs are not elite — **they are underpaid**. The employer is struggling because the salary is below market. And for a job seeker, that is not a warning sign — that is **negotiating power**. If you see a job that keeps being reposted, you are allowed to ask for more.

Part 4 — Handover (2:30 – 2:50)

So this page puts the whole market on one map — with the caveat on screen at all times: competition comes from the reliable window, postings up to June 2023.

But a map shows everyone the same picture. How do you turn it into advice for one specific person? That is the Career Recommender.

Numbers you must know cold

Figure	Value
The metric	applications ÷ seats = applicants per seat
Market average	3.8 applicants per seat
Lowest competition	Personal Care 0.7 · F&B 1.0
Highest competition	Social Services 11.3 · Risk Management 8.7
The spread	16x difference between categories
IT	140,866 postings · \$6,500 median · ~4.8 applicants/seat
Hard-to-fill pay gap	reposted \$3,700 vs first-time \$3,850
Share of ads reposted	4.1%
Competition window	postings up to 30 June 2023

Questions you own

"Is 'applicants per seat' a fair measure of competition?"

It's the most direct one this data allows: real applications divided by real seats. It has limits — one person can apply to many jobs — but it compares categories on equal terms.

"Why is the vertical axis logarithmic?"

Category sizes range from a few hundred postings to 140,000. On a normal axis, the small half of the market would be squashed flat against the bottom. We label the axis clearly.

"Doesn't low competition just mean nobody wants those jobs?"

Sometimes — that's why the map shows pay as colour at the same time. Low competition plus low pay is a warning. Low competition plus decent pay, like Personal Care at \$4,000, is an opportunity.

"Why do you trust the application counts at all?"

Only inside the reliable window — postings up to June 2023, where the counters had time to accumulate. That's about 200,000 postings, which is plenty.

Rehearsal checklist

- ☐ Timed at **3:00 or under**
- ☐ You can say "0.7, 1.0, 11.3 — a sixteen-fold spread" from memory
- ☐ You can explain the quadrants in two sentences
- ☐ The hard-to-fill twist delivered as a surprise, not read as a fact
- ☐ Handover line practised — it launches the demo